A Centralized Web Platform for Campus Event & Placement Management

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Certificate

This is to certify that the Practice School–II project work entitled **“A Centralized Web Platform for Campus Event & Placement Management”** submitted by **Ishaan Saraswat (2022BTECH042)** towards the partial fulfilment of the requirements for the degree of **Bachelor of Technology in Computer Science and Engineering** at **JK Lakshmipat University, Jaipur**, is a record of the work carried out by him under my supervision and guidance.

In my opinion, the submitted work has reached the level required for being accepted for the **Practice School–II Mid-Semester Examination**.

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ABSTRACT

The College Event & Placement Management System is a centralized web-based application developed to streamline and digitize campus event coordination and placement management processes. During the internship in the college administrative section, the entire system was conceptualized, designed, and developed independently to solve real-world administrative challenges.

Traditionally, colleges rely on WhatsApp groups, Google Forms, spreadsheets, and manual paperwork to manage events and placements. This creates confusion, data inconsistency, lack of transparency, and inefficiency. The proposed system acts as a single digital hub where students, event coordinators, placement cell members, and administrators can manage their responsibilities in a structured manner.

The platform follows a role-based architecture with four user roles: Student, Event Admin, Placement Cell, and Super Admin. The system ensures secure authentication, hierarchical event management, structured placement workflows, and real-time data tracking.

The application is built using React with TypeScript for frontend development, Tailwind CSS for responsive UI design, and Supabase for backend services including authentication, database management, and storage.

Chapter 1: ABOUT THE ORGANIZATION

JK Lakshmipat University (JKLU) is a private university located in **Jaipur, Rajasthan**, established in **2011** by the **JK Organisation**, one of India's leading industrial groups with a legacy of more than 135 years. The university is committed to providing quality education through innovation, research, and industry-oriented learning.

JKLU aims to develop future-ready professionals by combining strong academic foundations with practical exposure, interdisciplinary learning, and real-world problem solving. The university offers undergraduate, postgraduate, and doctoral programs in fields such as **Engineering, Management, and Design** through its different institutes including the **Institute of Engineering and Technology (IET)**, **Institute of Management**, and **Institute of Design**.

The campus provides modern infrastructure including advanced laboratories, digital learning facilities, research centers, and collaborative workspaces that support hands-on learning and innovation. JKLU emphasizes experiential learning through projects, internships, industry collaborations, and entrepreneurship initiatives.

With a focus on **innovation, leadership, and global exposure**, JK Lakshmipat University continues to create an academic environment that encourages creativity, critical thinking, and technological advancement among students.

Chapter 2: INTRODUCTION

In modern academic institutions, event management and placement coordination are critical operational components. However, these processes are often handled through fragmented tools, resulting in data duplication and administrative burden.

The College Event & Placement Management System was developed as a solution to centralize and automate these workflows. The system introduces structured data management, automated eligibility filtering, role-based dashboards, and transparent tracking mechanisms.

The core idea behind this project was to build a scalable and secure platform that can serve as the official digital hub of the campus.

Objectives of the System:

- Create a centralized digital platform for events and placements.
- Reduce dependency on manual forms and spreadsheets.
- Enable structured role-based access control.
- Improve transparency in placement shortlisting and selection.
- Provide detailed analytics and tracking features.

Chapter 3: TOOLS & TECH STACK USED

Frontend Technologies:
• React.js – Component-based UI development.
• TypeScript – Type safety and maintainable code.
• Tailwind CSS – Responsive and utility-first styling.
• shadcn/ui – Pre-built UI components for consistency.

Backend & Services:
• Supabase – PostgreSQL database, authentication, and storage.
• Row-Level Security (RLS) – Ensures role-based data protection.

Authentication:
• Email & password authentication.
• Automatic Super Admin role assignment for first user.
• Role-based protected routing system.

Database Structure:
• Users Table
• Events Table (Main Events)
• Sub-Events Table
• Registrations Table
• Companies Table
• Job Postings Table
• Applications Table
• Placement Results Table

Chapter 4: ROLES AND RESPONSIBILITIES

Internship Role in the Organization:

During the internship in the Administrative Section of the college, the intern independently performed the following roles while developing the system:

- Full Stack Developer – Designed and implemented frontend and backend.
- System Analyst – Conducted requirement gathering and workflow analysis.
- Database Designer – Structured relational database schema.
- UI/UX– Designed responsive dashboards for each role.
- Deployment & Testing– Conducted debugging and system testing.
- Documentation – Prepared system documentation and reports.

The project was developed independently without a development team, demonstrating ownership, problem-solving ability, and technical proficiency.

Chapter 5: WORK DONE

The internship was undertaken in the Administrative Section of the college, where the task was to independently conceptualize, design, and develop a fully functional College Event & Placement Management System. The entire development lifecycle — from requirement gathering to deployment-ready testing — was executed over a period of six weeks (15th January to 27th February). The work was structured into eight well-defined phases, each addressing a critical component of the system.

Phase 1: Requirement Analysis & Planning

The first phase involved in-depth discussions with the college's administrative staff, placement cell members, and event coordinators to understand the pain points of the existing system. The following activities were carried out:

- Conducted meetings with administrative staff to identify limitations of current manual processes such as WhatsApp groups, Google Forms, and spreadsheets.
- Studied and documented existing event registration and placement workflows to identify bottlenecks and data gaps.
- Prepared a comprehensive system blueprint listing all required features, user roles, and access permissions.
- Defined four distinct user roles: Student, Event Admin, Placement Cell, and Super Admin, with clear responsibility boundaries for each.
- Created a prioritized feature list and technology selection matrix, finalizing React, TypeScript, Tailwind CSS, and Supabase as the core tech stack.

Phase 2: System Architecture Design

A modular and scalable architecture was designed before writing any code. The key design decisions taken during this phase include:

- Designed a three-layer modular architecture: a React-based frontend layer, a Supabase-powered backend and database layer, and a security layer using Row-Level Security (RLS) policies.
- Planned role-based routing where each user type is directed to a dedicated protected dashboard immediately upon login.
- Designed a hierarchical event model where Main Events act as parent records and Sub-Events are child entries linked by foreign key relationships.
- Defined component structure for the frontend ensuring reusability, separation of concerns, and scalability for future module additions.
- Finalized data flow diagrams mapping how each role interacts with the system and what data they can read, write, or manage.

Phase 3: Database Design & Implementation

A normalized relational database schema was designed and implemented on Supabase. The database was structured to support complex queries, relationships, and role-specific data access.

- Designed and created eight core tables: Users, Events, Sub-Events, Registrations, Companies, Job Postings, Applications, and Placement Results.
- Established foreign key relationships between tables to maintain referential integrity — for example, Sub-Events linked to Events, Applications linked to Job Postings and Users.
- Added CHECK constraints and NOT NULL constraints on critical fields such as role, status, and eligibility criteria to prevent invalid data entry.
- Wrote and tested Row-Level Security (RLS) policies for each table so that data access is automatically restricted based on the authenticated user's role, preventing unauthorized cross-role data reads or writes.
- Configured Supabase storage buckets for handling event photo uploads and placement-related document storage with appropriate access controls.

Phase 4: Authentication & Security Setup

A secure and role-aware authentication system was configured to ensure that every user receives a session-level identity and is routed to the correct interface. The following work was done in this phase:

- Configured Supabase email and password authentication with secure session management and token-based access control.
- Implemented automatic Super Admin role assignment logic: the very first registered user is automatically granted the Super Admin role, removing the need for manual database intervention.
- Built a role-based protected routing system in React that reads the authenticated user's role from the database and redirects them to their respective dashboard, blocking unauthorized route access.
- Applied Supabase Row-Level Security policies at the database level as a second layer of protection ensuring even if the frontend routing is bypassed, data cannot be accessed without the correct role.

Phase 5: Frontend Development (Role-Based Dashboards)

Four dedicated dashboards were developed using React with TypeScript and styled using Tailwind CSS and shadcn/ui component library. Each dashboard was tailored to the specific needs of its user role:

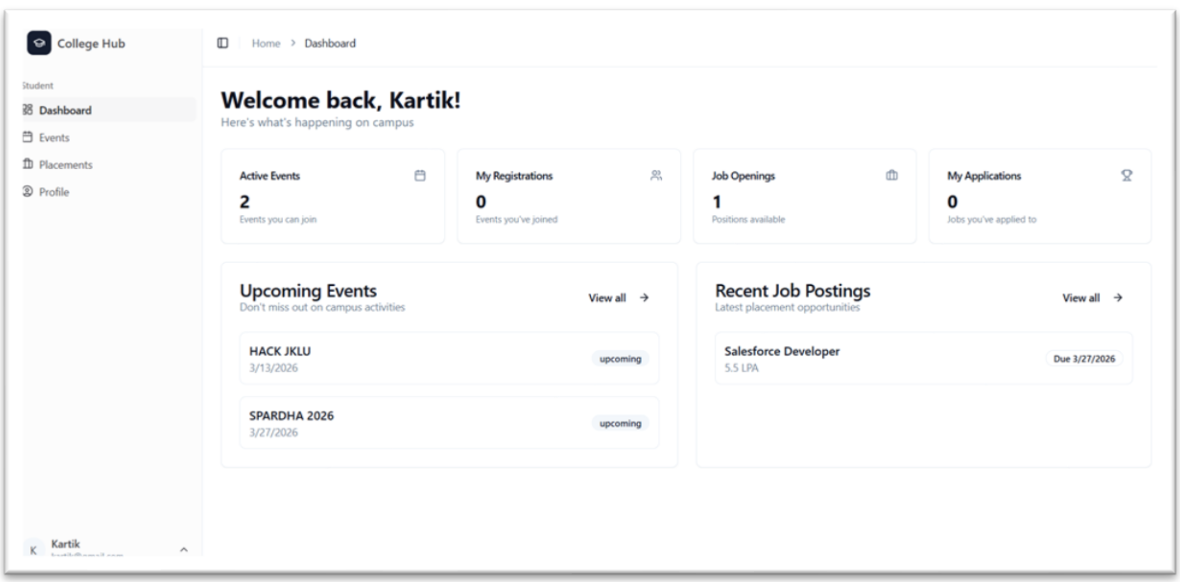
• **Student Dashboard:** Displays available events and sub-events, allows registration with one click, tracks registration status, shows applied job postings, and displays application and placement outcomes.

• **Event Admin Dashboard:** Enables creation and management of Main Events and Sub-Events, setting participant limits, viewing registrations, publishing results, and uploading event photos to the gallery.

• **Placement Cell Dashboard:** Provides tools to add and manage companies, create job postings with eligibility criteria (CGPA, branch, backlogs), review student applications, and record offer letters and salary packages.

• **Super Admin Dashboard:** Provides full system oversight including user management, role assignment, system-wide data visibility, and the ability to configure or override settings across all modules.

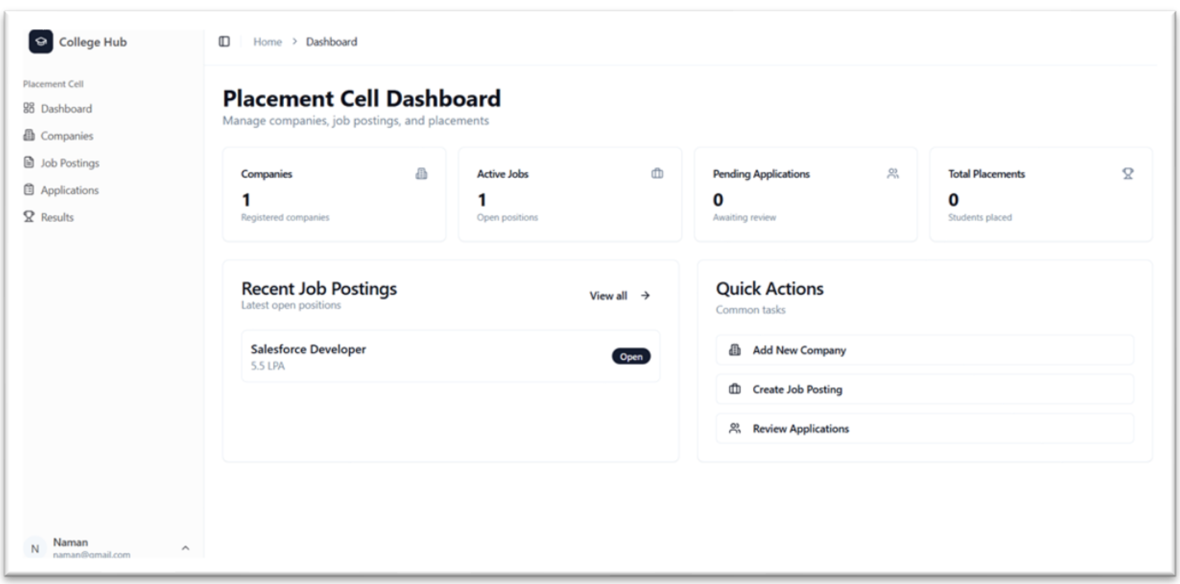
• Implemented dynamic forms with field-level validation, real-time filtering logic (e.g., eligible students filtered automatically for a job), and live status tracking components across all dashboards.



Phase 6: Placement Module Development

The placement module is the most data-intensive component of the system and was developed with end-to-end workflow management. This phase covered the full lifecycle from company onboarding to final placement result recording:

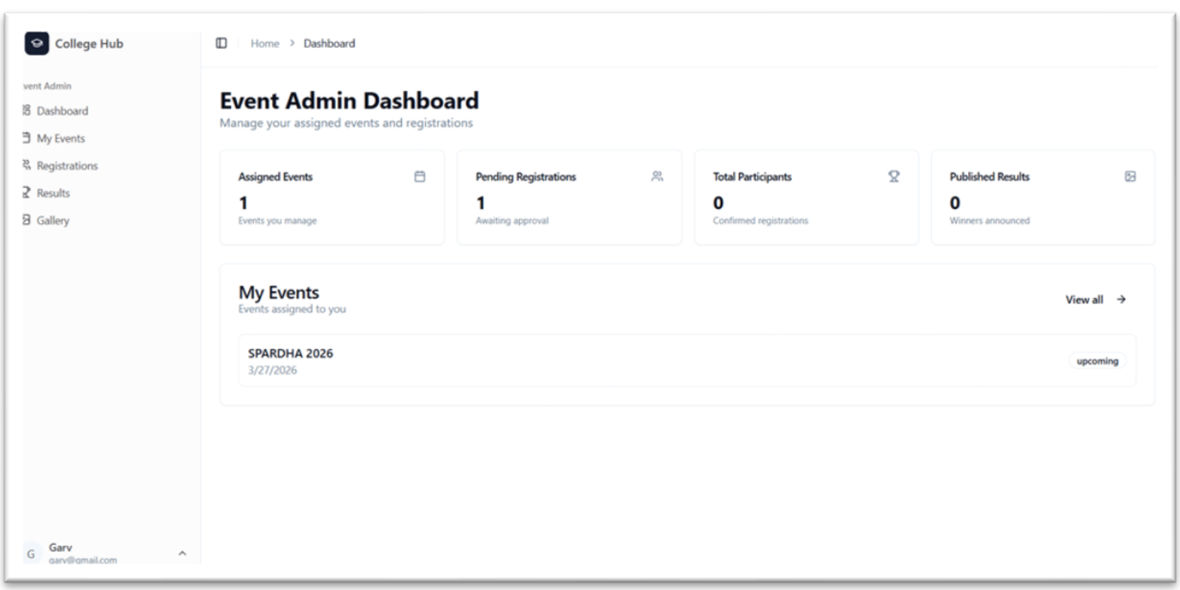
- **Company Management System:** Built a CRUD interface for the placement cell to add, update, and manage company records including company name, industry type, website, and contact details.
- **Job Posting with Eligibility Filters:** Developed a job posting creation form allowing placement cell to specify eligibility parameters such as minimum CGPA, allowed branches, maximum allowed backlogs, and batch year. The system automatically filters out ineligible students from applying.
- **Application Review Dashboard:** Built a tabular interface displaying all student applications for a job posting with filtering, sorting, and status update controls. The placement cell can mark applications as Shortlisted, Under Review, or Rejected.
- **Offer & Package Recording Module:** Developed a result entry form where placement cell members can record offer letter details, CTC package, joining date, and location for selected students. These results are reflected on the student's own dashboard in real time.



Phase 7: Event Module Development

The event module supports complex multi-level event structures, participant management, result publishing, and media storage. The following features were developed:

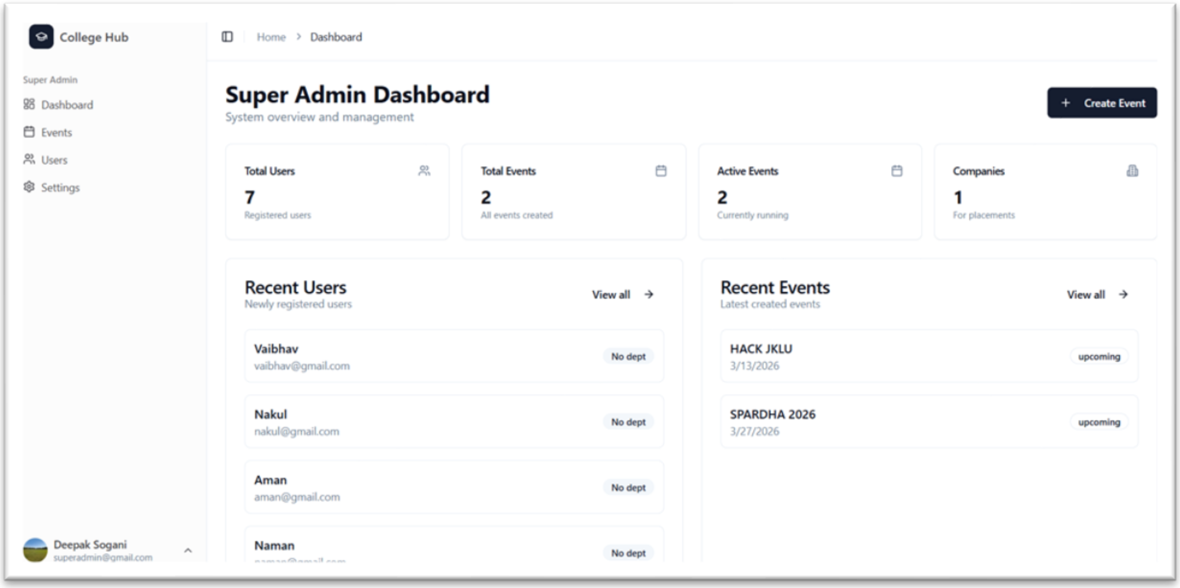
- **Main Event Creation:** Event Admins can create Main Events with title, description, date, venue, and category. Each event acts as a parent record under which sub-events are organized.
- **Sub-Event Management:** Developed a sub-event interface where each Main Event can have multiple sub-events (e.g., Group Dance, Solo Singing, Debate). Each sub-event has its own registration capacity, date, and status.
- **Participant Limit Logic:** Implemented automatic capacity enforcement where the registration button is disabled or hidden once a sub-event reaches its maximum participant limit, preventing over-registration without manual intervention.
- **Result Publishing System:** Built a result entry interface for Event Admins to publish sub-event results (1st, 2nd, 3rd place winners) which are then visible to all students on their dashboard and event listing pages.
- **Photo Gallery Integration:** Integrated Supabase Storage for photo uploads tied to each event. Event Admins can upload event photos which are rendered in a responsive gallery view accessible to all users, creating a digital archive of college events.



Phase 8: Testing & Optimization

Before finalizing the system, thorough manual testing was conducted across all four user roles to validate functionality, security, and user experience. Key testing and improvement activities included:

- Role-Based Testing: Logged in as each of the four user types and performed all available actions to verify correct functionality and access restrictions.
- Security Testing: Attempted to access unauthorized routes and data directly via URL or API calls to confirm RLS policies and protected routing are working as intended.
- Edge Case Validation: Tested boundary conditions such as registering for a full sub-event, applying to a job with ineligible profile data, and submitting incomplete forms to confirm proper validation messages.
- Bug Fixing: Identified and resolved issues related to data filtering, form submission, routing edge cases, and UI rendering inconsistencies across different screen sizes.
- UI/UX Improvements: Refined component layouts, spacing, font hierarchy, and color consistency to improve readability and visual clarity across all dashboards.
- Performance Optimization: Minimized unnecessary re-renders in React, optimized Supabase queries with appropriate select filters, and ensured fast page load times by lazy-loading non-critical components.



Week-Wise Logbook (15th January – 27th February)

Week 1 (15 Jan – 21 Jan): Requirement Gathering & Planning

- Met with administrative staff and placement cell members to understand existing challenges and manual workflows.
- Studied and documented event registration and placement workflows used currently (WhatsApp, Google Forms, spreadsheets).
- Defined user roles (Student, Event Admin, Placement Cell, Super Admin) and their access permissions.
- Prepared a detailed system blueprint with feature list, tech stack selection, and a 6-week development plan.

Week 2 (22 Jan – 28 Jan): Database Design & Authentication

- Designed the full relational database schema with all 8 tables, foreign key relationships, and field constraints.
- Created the Supabase project, set up the PostgreSQL database, and implemented all table schemas.
- Configured Supabase email/password authentication and set up the automatic Super Admin assignment logic for the first registered user.

- Wrote and tested Row-Level Security (RLS) policies for all tables to enforce role-based data access at the database level.

Week 3 (29 Jan – 4 Feb): Student Dashboard & Event Browsing

- Set up the React + TypeScript + Tailwind CSS project structure with protected routing for all four user roles.
- Developed the Student dashboard with event listing, sub-event browsing, and one-click registration with capacity enforcement.
- Implemented real-time registration status tracking on the student's dashboard showing registered sub-events and their outcomes.
- Built the event photo gallery view accessible to all students, fetching images stored in Supabase storage.

Week 4 (5 Feb – 11 Feb): Event Admin Dashboard & Event Module

- Developed the Event Admin dashboard with full CRUD capabilities for Main Events and Sub-Events.
- Implemented the participant limit enforcement system that automatically closes sub-event registrations when the capacity is reached.
- Built the result publishing module where Event Admins can enter and publish winners for each sub-event.
- Integrated the photo upload feature using Supabase storage, allowing Event Admins to upload and manage event photos per event.

Week 5 (12 Feb – 18 Feb): Placement Cell Dashboard & Placement Module

- Developed the Placement Cell dashboard with company management, job posting creation, and application review features.
- Implemented the eligibility filtering logic that automatically validates student profiles against job eligibility criteria (CGPA, branch, backlogs) before allowing an application.
- Built the application review interface with status management (Shortlisted, Under Review, Rejected) and batch update controls.
- Developed the offer and package recording module for entering final placement results visible on student dashboards in real time.

Week 6 (19 Feb – 27 Feb): Super Admin, Testing & Final Documentation

- Completed the Super Admin dashboard with full user management, role assignment interface, and system-wide data overview.
- Implemented and verified protected routing for all roles, confirming that unauthorized route access is blocked correctly.
- Conducted full end-to-end system testing across all four roles, fixed identified bugs, and refined UI/UX consistency.

• Performed security testing to validate RLS policy enforcement, checking that cross-role data access is prevented at the API level.

• Optimized React component rendering and Supabase query efficiency for improved application performance.

• Prepared final system documentation, internship report, and project summary for submission.

Chapter 6: FUTURE PLAN

1. AI-Based Resume Parsing and Automatic Shortlisting The system currently relies on manual eligibility filters set by the placement cell. In future iterations, an AI/ML module can be integrated that automatically parses uploaded student resumes, extracts key fields such as skills, certifications, projects, and experience, and matches them against job requirements. This would significantly reduce manual screening effort and improve shortlisting accuracy.

2. Real-Time Notifications via Email and SMS Currently, students must log in to check updates on event results, application status, or new job postings. A notification service using tools like Twilio (for SMS) and SendGrid or Nodemailer (for email) can be integrated so that students and admins receive instant alerts for key events — such as registration confirmation, shortlisting updates, result announcements, and new placement drives.

3. Advanced Analytics Dashboard for Placement Statistics A dedicated analytics module can be developed to provide visual insights into placement data — including year-wise placement trends, branch-wise offer statistics, average and highest CTC packages, company visit frequency, and student participation rates in events. This would help the institution make data-driven decisions and present placement records to external stakeholders effectively.

4. Mobile Application Version To improve accessibility, a mobile application version of the platform can be developed using React Native, leveraging the existing Supabase backend without requiring any backend changes. This would allow students and admins to manage registrations, check results, and track placement status directly from their smartphones, making the platform accessible beyond the campus network.

5. Integration with College ERP System The platform currently operates as a standalone system. In the future, it can be integrated with the college's existing ERP system to auto-import student academic data — such as CGPA, attendance, and branch details — eliminating the need for manual profile updates and ensuring placement eligibility criteria are always based on accurate, up-to-date academic records.

Chapter 7: CONCLUSION

The College Event & Placement Management System represents a significant step toward the digital transformation of campus administration. Developed independently over the course of a six-week internship in the college's administrative section, the project addresses a real and persistent problem — the fragmented, manual, and error-prone nature of managing events and placements across a large student body.

The system successfully consolidates what previously required multiple disconnected tools — WhatsApp groups, Google Forms, spreadsheets, and manual record-keeping — into a single, structured, and secure web platform. By introducing role-based access control, automated eligibility filtering, hierarchical event management, and real-time data tracking, the platform eliminates redundancy and brings transparency to processes that previously lacked it.

From a technical standpoint, the project demonstrates hands-on proficiency across the full development lifecycle — from requirement analysis and database design to frontend development, security implementation, and end-to-end testing. The use of modern technologies such as React with TypeScript, Tailwind CSS, and Supabase reflects an understanding of current industry practices and the importance of building scalable, maintainable systems.

Beyond technical skills, this internship reinforced the value of independent problem-solving, structured project planning, and the ability to translate real-world administrative challenges into functional software solutions. Every design decision — from the database schema to the RLS policies — was taken with long-term scalability and institutional usability in mind.

The platform is deployment-ready and built to scale. With the planned future enhancements — including AI-based shortlisting, real-time notifications, mobile access, and ERP integration — the system has the potential to become the central digital backbone of the college's administrative operations. It stands as a practical demonstration that thoughtful software development can meaningfully improve institutional efficiency and the overall student experience.